Research Note: Transport Bounds for Conditional and Path-Dependent Event Contracts

Derivative Parity Research — working manuscript

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Abstract

We formulate finite-state, model-robust bounds for binary, conditional, and path-dependent event contracts using linear programming, multi-marginal optimal transport, and (where the event depends on a dynamically traded asset) martingale optimal transport. The distinction between static consistency and dynamic self-financing hedging is central. Three reproducible *synthetic* examples yield: a three-event intersection price in $[0.15, 0.22]$; a rally-then-decline price in $[1/12, 1/4]$ under martingale constraints, versus $[0, 1/2]$ under ordinary couplings; and a three-date rally-decline-rebound price in $[0, 1/16]$ versus $[0, 1/4]$. These numerical examples are demonstrations rather than claims of new theoretical results. We state a falsifiable empirical research agenda for assessing interval contraction and executable violations using settlement-aligned cross-market data.

1 Economic interpretation and identification

A unit-collateralized binary claim has payoff $H\colon \Omega \to \{0,1\}$, where Ω is an explicit, common set of settlement states. If all contracts share a numeraire, settlement time and consistent pricing measure Q, their normalized present values are $\mathbb{E}_Q[H]$. These are *pricing-measure* expectations, not automatically physical probabilities. Differences in oracle criteria, observation windows, collateral, fees, settlement delays and tradeability invalidate naive cross-market parity. A static price interval is a *consistency bound* under the hypotheses, not automatically a directly executable arbitrage.

Conditional-token positions can be intersections of distinct conditions, but a position paying on $A \cap B$ is not itself a normalized conditional claim paying $\mathbf{1}_B$ given A. Where $Q(A) > 0$, $Q(B \mid A) = Q(A \cap B)/Q(A)$, with careful treatment of the conditional settlement mechanism (Conditional Tokens contributors, 2026). Prices $Q(B \mid A)$ and $Q(A \cap B)$ cannot be interchanged.

2 Finite-state transport and its dual

Let $\Omega = \{\omega_1, \dots, \omega_N\}$, $q_i = Q(\omega_i)$, and let $h_j(\omega)$ be observed contract payoffs. With observed executable or modeling quote intervals $[\ell_j, u_j]$, consider

$$\mathcal{Q} = \big\{ q \in \mathbb{R}^N_+ : \mathbf{1}^\top q = 1, \quad \ell_j \le h_j^\top q \le u_j \ (j = 1, \dots, m), \tag{1}$$

$$Dq = 0 \big\}, \tag{2}$$

where D optionally encodes structural restrictions. The sharp robust interval for target payoff H is

$$\underline{\pi}(H) = \min_{q \in \mathcal{Q}} H^\top q, \qquad \overline{\pi}(H) = \max_{q \in \mathcal{Q}} H^\top q. \tag{3}$$

For exact quotes $h_j^\top q = p_j$ and $D = 0$, finite-dimensional LP duality yields

$$\overline{\pi}(H) = \min_{a_0,a} \quad a_0 + \sum_j a_j p_j \quad \text{s.t.} \quad a_0 + \sum_j a_j h_j(\omega) \geq H(\omega), \quad \forall \omega. \tag{4}$$

The lower dual replaces "min" by "max" and "$\geq$" by "$\leq$". A dual solution is a statewise sub/super-replicating portfolio. Inequality quotes require the complete inequality dual; one cannot simply reuse the exact-quote hedge coefficients. General theory is developed by Beiglbock et al. (2013).

3 Example I: joint and conditional binary claims

Let A, B, C be binary outcomes observed at three monitoring dates and assume the following *hypothetical, normalized* price constraints:

$$(p_A, p_B, p_C, p_{AB}, p_{BC}, p_{AC}) = (.60, .55, .45, .40, .30, .22). \tag{5}$$

Let $q = Q(A \cap B \cap C)$. Inclusion-exclusion and monotonicity imply

$$0.15 = p_{AB} + p_{BC} - p_B \leq q \leq p_{AC} = 0.22. \tag{6}$$

These bounds are sharp: the attached LP produces feasible eight-state couplings at both endpoints and certifies the dual inequalities

$$\mathbf{1}_{AB} + \mathbf{1}_{BC} - \mathbf{1}_B \leq \mathbf{1}_{ABC} \leq \mathbf{1}_{AC}. \tag{7}$$

The observed $p_{AB} = 0.40$ fixes the conditioning denominator, hence $Q(C \mid A \cap B) \in [0.375, 0.55]$. With only the three marginals, the interval for $Q(ABC)$ is $[0, 0.45]$. With AB and BC added, it shrinks to $[0.15, 0.30]$.

4 Example II: a two-date price-path digital

Here the state is a *tradable*, appropriately discounted asset price, with zero interest, payouts and funding adjustments in the synthetic model. Prescribe

$$S_1 \in \{90, 110\}, \qquad Q(S_1 = 90) = Q(S_1 = 110) = \tfrac{1}{2}, \tag{8}$$

$$S_2 \in \{70, 90, 110, 130\}, \qquad Q(S_2 = s) = \tfrac{1}{4} \quad \text{for each } s. \tag{9}$$

The target is a monitored rally followed by a decline:

$$H = \mathbf{1}_{\{S_1 = 110\}} \mathbf{1}_{\{S_2 \leq 90\}}. \tag{10}$$

Ordinary OT over couplings of the two marginals yields $[0, 1/2]$. The martingale restriction $\mathbb{E}_Q[S_2 \mid S_1] = S_1$ leads to a smaller interval.

Proposition 1. *For the two marginals above, among martingale couplings the sharp prices of H are $\underline{\pi}(H) = 1/12$ and $\overline{\pi}(H) = 1/4$.*

Proof. Write $u = Q(110, 70)$, $v = Q(110, 90)$, $w = Q(110, 110)$ and $z = Q(110, 130)$. The row sum and conditional martingale equation are

$$u + v + w + z = \tfrac{1}{2}, \qquad -40u - 20v + 20z = 0. \tag{11}$$

Thus $z = 2u + v$ and $w = \frac{1}{2} - 3u - 2v$. Each column has mass $1/4$, so $z \le 1/4$ and $w \le 1/4$. Consequently $u + v \le z \le 1/4$ and $3(u + v) \ge 3u + 2v \ge 1/4$, yielding $1/12 \le u + v \le 1/4$. For the lower endpoint, take the conditional-row masses $(u, v, w, z) = (1/12, 0, 1/4, 1/6)$; for the upper endpoint take $(0, 1/4, 0, 1/4)$. Subtracting these rows from the respective column marginals yields nonnegative complementary rows. Since the unconditional mean of S_2 equals that of S_1, the remaining row is a martingale too. Both bounds are attained. □

Proposition 2 (Explicit sub- and superhedges)**.** *Let $G = \mathbf{1}_{\{S_1=110\}}(S_2 - S_1)$ be the gain on a predictable self-financing position in the discounted underlying. Then for every permitted pair (S_1, S_2),*

$$\underbrace{\tfrac{1}{3}\mathbf{1}_{\{S_1=110\}} - \tfrac{1}{3}\mathbf{1}_{\{S_2=110\}} - \tfrac{1}{60}G}_{L} \le H \le \underbrace{\mathbf{1}_{\{S_2=130\}} - \tfrac{1}{20}G}_{U}. \tag{12}$$

The costs of L and U are $1/12$ and $1/4$, respectively.

Proof. Check the eight possible (S_1, S_2) states. The martingale condition implies $\mathbb{E}_Q[G] = 0$, while the observed marginals price the digital terms at $1/2$ and $1/4$. Equality of the optimal values follows from the extremal couplings above. The test suite checks both pointwise inequalities and all equality constraints. □

A further *hypothetical* joint quote $Q(S_1 = 110, S_2 = 130) = 0.20$ shrinks the martingale interval to $[0.10, 0.20]$. This quote is a modeling input in the toy experiment, not observed data.

5 Conditional ratios when the conditioning price is uncertain

If $Q(A)$ is not a point quote but falls in $[\ell_A, u_A]$, the ratio $Q(A \cap B)/Q(A)$ cannot always be bounded by dividing unrelated marginal interval endpoints: the feasible numerator and denominator may be dependent. One can instead optimize

$$\min_{q \in \mathcal{Q},\, a^\top q \ge \delta} \frac{(ab)^\top q}{a^\top q}, \qquad \max_{q \in \mathcal{Q},\, a^\top q \ge \delta} \frac{(ab)^\top q}{a^\top q}, \quad \delta > 0, \tag{13}$$

using the Charnes–Cooper variables $y = q/(a^\top q)$ and $t = 1/(a^\top q)$. Then $a^\top y = 1$, $\mathbf{1}^\top y = t$ and $\ell_j t \le h_j^\top y \le u_j t$ become linear constraints. This gives sharp fractional conditional bounds with market quote bands, subject to the same common-state and positive-denominator assumptions. The accompanying code implements the transformation and unit tests.

6 Example III: the full three-date filtration matters

Take $S_1 \in \{90, 110\}$ and $S_2 \in \{70, 90, 110, 130\}$ with the preceding marginals, plus

$$S_3 \in \{50, 70, 90, 110, 130, 150\}, \qquad \mu_3 = (.125, .125, .25, .25, .125, .125). \tag{14}$$

For a three-date claim

$$H_3 = \mathbf{1}_{\{S_1=110,\, S_2 \le 90,\, S_3 \ge 130\}}, \tag{15}$$

the 48-atom LP returns ordinary OT bounds $[0, 1/4]$, whereas full-filtration MOT bounds are $[0, 1/16]$. To enforce a martingale *in the natural price filtration*, not merely in its last observed state, we require for every observed history $s_{1:t}$,

$$\sum_{\omega \in \Omega} q(\omega)\, \mathbf{1}_{\{S_{1:t} = s_{1:t}\}}(S_{t+1} - S_t) = 0, \quad t = 1, 2. \tag{16}$$

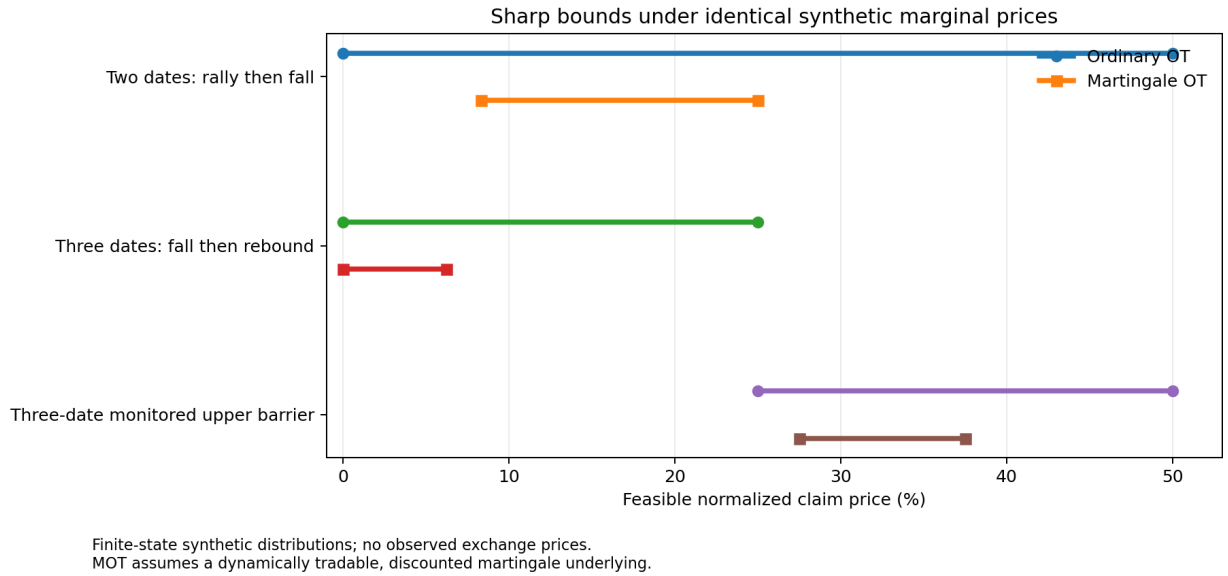


Figure 1: Sharp feasible intervals from synthetic marginal distributions. The martingale restriction changes the set of admissible path laws and requires an appropriately discounted, dynamically traded underlying.

The code verifies primal feasibility, pointwise dual certificates and numerical attainment at each endpoint. The inequality is a discrete monitoring result only; continuously monitored crossings need a different state space and additional assumptions. For comparison, the price of touching 130 at any of the *three* monitoring dates is $[0.25, 0.50]$ under ordinary OT and $[0.275, 0.375]$ under full-filtration MOT.

7 Research scope and non-novelty guardrails

Martingale OT and robust path-dependent option pricing are established (Beiglbock et al., 2013; Guo and Loeper, 2021; Engstrom et al., 2025). Adapted transport and robust multicausal hedging have also been studied (Krsek and Pammer, 2025). Recent prediction-market work already treats logical combinatorial arbitrage (Shi and Zhai, 2026) and coherent joint parlay market making (Rana et al., 2026). Accordingly, neither finite-state linear programming nor copula bounds constitutes a standalone novelty claim. In the prediction-market setting, temporal arrival of information should not be confused with an adapted-transport constraint: the latter must specify *which processes are coupled and what information their coupling is allowed to use.*

8 Falsifiable empirical agenda (not yet executed)

1. Construct a settlement-semantics audit: oracle, event definition, cutoff, observation schedule, currency, collateral, settlement lag.

2. Recover contract payoff vectors on compatible joint states; store bid/ask/depth snapshots *with timestamps*, not last-trade prices.

3. Compute successive intervals: singleton marginals, pairwise joints, conditional claims, and (only for traded underlying) option-calibrated martingale constraints.

4. Record interval width, interval contraction, model feasibility, and executable bound violations after spread, size, fees and timing costs.

5. Hold out joint contracts for out-of-sample interval coverage and compare entropy-regularized priors against maximum-entropy and parametric copula baselines. Report lack of identification when the interval remains wide.

No live order books or venue-level claims have been analyzed in this note. All market inputs and numeric results are synthetic. The package is research software, not a trading system or investment recommendation.

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